

Federated Edge Intelligence for Enhanced Security in Consumer Intermittent Healthcare Devices Using Adversarial Examples

Farhan Ullah^{id}, Leonardo Mostarda^{id}, Diletta Cacciagrano^{id}, Mohammed J. F. Alenazi^{id}, *Senior Member, IEEE*, Chien-Ming Chen^{id}, *Senior Member, IEEE*, and Saru Kumari^{id}, *Senior Member, IEEE*

Abstract—Modern consumer electronics that integrate the concept of connected personal medical devices are known as Smart Healthcare Systems (SHS). The SHS utilized healthcare devices and edge computing for data capture, transmission via smart devices, analysis, and the delivery of healthcare services. Cyberattacks on consumer medical devices in the healthcare sector are becoming increasingly common. Technological frameworks like edge computing can act as an intermediary layer between the cloud and SHS, reducing the burden and enhancing data security and privacy. In this study, we proposed a novel edge intelligence approach for improving medical device security that employs Federated Learning (FL) and network traffic adversarial examples. First, we generated a distinctive dataset using image-based features extracted from network traffic bytes. Secondly, due to the intermittent behavior of clients, the dataset received from each client may be imbalanced, which can negatively affect performance. Third, adversarial examples are generated to assess the robustness of datasets, utilizing four distinct types of adversarial attack methods that introduce perturbations into the input data. Finally, the cooperative FL architecture ensured data security, privacy, and edge intelligence. The proposed method is analyzed using two standard datasets, CICIoMT2024 and Edge-IIoT, achieving 97.45% and 96.7% detection accuracy, respectively.

Index Terms—Consumer healthcare devices, security and privacy, edge computing, federated learning, adversarial attacks, network traffic.

I. INTRODUCTION

THE HEALTHCARE industry is undergoing a paradigm shift due to integrating modern consumer electronics and technologies with medical devices to enhance patient

care and outcomes. The Internet of Medical Things (IoMT), which consists of a vast network of interconnected medical devices, has substantially revolutionized healthcare delivery. These devices have a variety of applications, including electronic health record management, medicine administration systems, diagnostics, and patient monitoring. However, the interconnection of these devices poses significant security and privacy concerns. The potential implications of security flaws in healthcare devices are severe, including unauthorized access to personal patient information and tampering with life-threatening medical equipment. Furthermore, the increasing sophistication of cyberattacks makes it harder to maintain the integrity and confidentiality of healthcare data. Medical devices can be vulnerable to various attacks, including ransomware, which encrypts data and demands payment to decrypt it. Unauthorized access occurs when hackers acquire control, modify settings, or steal data. Data theft refers to getting sensitive patient information. Additional risks include exploiting software or firmware flaws and changing features to cause harm or malfunction. Denial of service (DoS) attacks interrupt device functionality. Furthermore, adversaries can physically manipulate devices, access manufacturing processes (also known as supply chain assaults), and monitor communication using man-in-the-middle attacks. Social engineering tactics can be used to trick individuals into granting unauthorized access or distributing malware [1], [2], [3]. The need to analyze enormous volumes of data collected daily to form inferences has led to several advances in machine learning and deep learning. Nonetheless, due to the immense value of this information, it is vital to safeguard the privacy and security of the examined data. In certain circumstances, regulatory compliance demands adopting protections to protect sensitive information, such as the General Data Protection Regulation (GDPR) [1].

Figure 1 depicts the purpose of an adversarial attack, which is to disrupt or destroy a system, obtain unauthorized access, or defeat security measures by leveraging vulnerabilities. The model training and prediction generation process requires transmitting individual client data to a centralized server. The resulting models are then distributed to each client. The fundamental risk associated with this technique is the insufficient security of confidential data. The interception of data transfers between consumers and the central server is a potential vulnerability. Furthermore, keeping massive amounts of data requires high-bandwidth, low-latency communication to make

Received 26 June 2024; revised 23 August 2024, 8 October 2024, 16 October 2024, and 2 December 2024; accepted 2 December 2024. Date of publication 4 December 2024; date of current version 14 August 2025. This work was supported by the Researcher Supporting Project, King Saud University, Riyadh, Saudi Arabia, under Grant RSPD2024R582. (Corresponding author: Saru Kumari.)

Farhan Ullah and Diletta Cacciagrano are with the Division of Computer Science, University of Camerino, 62032 Camerino, Italy (e-mail: farhan.ullah@unicam.it; diletta.cacciagrano@unicam.it).

Leonardo Mostarda is with the Department of Mathematics and Computer Science, University of Perugia, 06123 Perugia, Italy (e-mail: leonardo.mostarda@unipg.it).

Mohammed J. F. Alenazi is with the Department of Computer Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 11451, Saudi Arabia (e-mail: mjalenzi@ksu.edu.sa).

Chien-Ming Chen is with the School of Artificial Intelligence (School of Future Technology), Nanjing University of Information Science and Technology, Nanjing 211544, China (e-mail: chienmingchen@ieee.org).

Saru Kumari is with the Department of Mathematics, Chaudhary Charan Singh University, Meerut 250004, India (e-mail: saryusirohi@gmail.com).

Digital Object Identifier 10.1109/TCE.2024.3511615

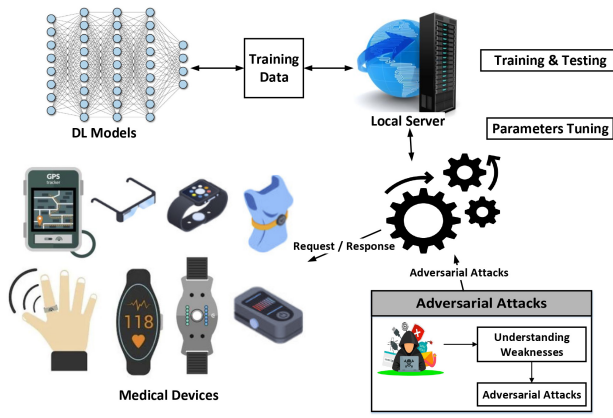


Fig. 1. Adversarial attacks on conventional model training process.

accurate forecasts. One possible solution to these concerns would be to centrally train the machine learning model by combining initial data from multiple clients. The trained model is then replicated and disseminated to each client, removing the need to transmit data for periodic updates. This methodology has two significant advantages over the traditional approach. First, minimizing network dependencies and latency enables clients to make predictions locally. Furthermore, it lowers communication overhead. This method requires initial data transfer from clients to the central server during training, which may jeopardize sensitive information. Furthermore, data privacy concerns may deteriorate due to large data transfers in scenarios such as IoT or Edge computing with low network quality [4].

FL is gaining popularity because it stores data on each local server, which is important for healthcare applications. It tackles privacy problems with distributed data analysis by storing confidential patient data on local computers and only exchanging updated models, not raw data. This decentralized strategy decreases the probability of adversarial attacks by removing a central repository that attackers can target. FL improves resilience by securely combining model weights from various sources, reducing the impact of adverse changes [5], [6]. The use of iterative model updates with aggregated weights improves predictive model dependability and accuracy while maintaining data confidentiality. This makes FL an effective strategy for dealing with hostile threats and protecting healthcare data and devices. The following are the main contributions of this study.

- 1) Image-based features are particularly useful for capturing structural data types such as storage, processes, and headers. We generated a dataset by crawling large-scale network traffic bytes. Following this, we developed an algorithm for converting network data into greyscale images.
- 2) To address overfitting and misclassification, we applied data augmentation techniques to balance images within each class. To enhance our dataset against adversarial attacks, we used strategies such as FGSM, IGSM, PGD, and a hybrid strategy that combined IGSM and PGD. This comprehensive approach increases the potential of the dataset to detect and protect against such threats while also improving data security.

- 3) We present a cooperative FL framework for detecting adversarial attacks on healthcare devices and demonstrate intelligent edge data processing. Through the utilization of shared data, our approach enhances security by preventing sensitive information from being disclosed.
- 4) Client numbers can vary dynamically, affecting classification accuracy. Despite the use of decentralized data, intermittent clients, and adversarial examples, the proposed strategy is more robust than centralized solutions.

The remaining part of the paper is organized as follows: Section II explains the related work, and Section III describes the proposed method. Section IV presents the experimental results, and Section V concludes the work.

II. RELATED WORK

Intrusion detection in healthcare devices gradually shifts from traditional machine learning-based methods to edge-based FL approaches [2], [7]. In this section, we discuss several methods involving machines and FL that detect and classify attacks on healthcare devices.

A. Machine Learning

Newaz et al. [8] demonstrated the susceptibility of personal medical device communications to several cyber threats. An adversary with malicious intent can intercept sensitive health data and execute denial-of-service, man-in-the-middle, replay, and fraudulent data injection attacks. Moreover, they introduced HEKA, which functions as an intrusion detection system (IDS). HEKA uses an n-gram-based technique and machine learning technologies to track and detect aberrant traffic trends in portable medical device interactions. HEKA achieves an F1-score of 98% and a detection accuracy of 98.4% when examined on eight publicly available devices. Fan et al. [9] explored the difficulty of constructing transferable adversarial scenarios for closed mode assaults without relying on the replacement model. This method used data-modal conversion to combine image-to-graph conversion, graph perturbation, and graph-to-image inversion. The solution outperforms the three baseline approaches in experiments and applies to real-world scenarios because it does not require prior knowledge of the victim model.

Zhang et al. [10] developed the DroidSIFT approach, which identifies malicious Android apps using relationship graphs and semantic information. The major goal of this system is to detect and prevent evolution attacks, emerging threats, and harmful variants. After evaluating 13,500 benign samples and 2,200 hazardous samples, DroidSIFT detected threats with an amazing 93% accuracy. Gao et al. [11] designed apps that use APIs for triggering connections and activities. These apps build a graph, which is then used in a network to produce embeddings for app categorization. GDroid employed graph neural networks to detect fraudulent apps as a security solution. The system accurately identified 98.99% of Android malware instances with a false positive rate below 1%. An efficient method for finding neural networks designed for Multi-Target Classification (MTC) was proposed

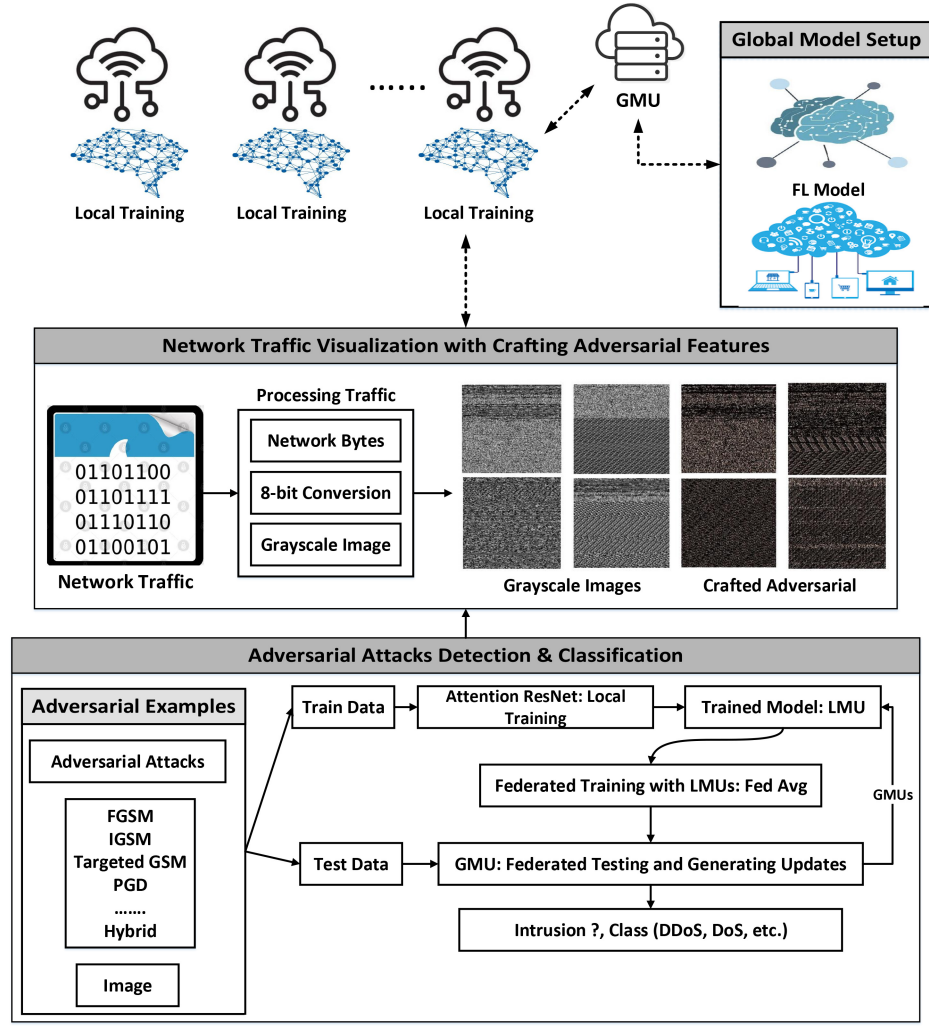


Fig. 2. Federated-based edge intelligence for healthcare devices using adversarial network traffic patterns.

by Zhang et al. [12]. This method determined the optimal model topologies by analyzing real-world Android network traffic. This method takes a distinctive search space and discrete design components into account to solve the problem as a limited optimization problem. This approach found appropriate MTC classification models by using the USTC-TFC2016 dataset.

B. Federated Learning Analysis

Several research studies [13], [14] used FL approaches in the healthcare domain. Schneble and Thamilarasu [15] described a massively distributed, machine learning-based IDS for Medical Cyber-Physical Systems (MCPS). It explored FL to reduce communication and compute costs in machine learning. Real patient data and security risks like DoS are used to test. The results showed a decrease in network communication costs and a 99.0% accuracy. The system used mobile device resources to achieve scalability and manage anomalous data distribution. Rehman et al. [16] described an FL framework for detecting side-channel threats on confidential medical data. Data was acquired from ten Android smartphone soft keyboard

users. The architecture trained a Deep Neural Network (DNN) model by training local models on two clients and then combining their updates. The DNN model demonstrated an accuracy of 80.09% following three validations conducted by each client. This result suggested that the model can detect side-channel attacks accurately. Wang et al. [17] used certificate-less signatures in FL to hide where parameter updates come from to protect medical devices from source inference attacks (SIAs). The proposed strategy incorporated an enhanced batch verification technique, which reduced the server's workload. Experimental results confirmed the efficacy of the suggested approach in lowering SIA success rates and accelerating signature verification. Mothukuri et al. [18] used an FL-based anomaly detection approach that uses data from individual devices to detect intrusions in IoT networks proactively. It maintains local data, uses federated training cycles with GRUs, and only communicates learned weights to the central server. The empirical findings indicate that decentralized systems exhibit better results in terms of privacy preservation and accuracy.

The main challenges in developing an FL technique are intermittent clients and imbalanced datasets. The primary

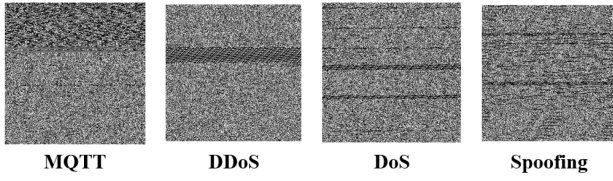


Fig. 3. Image extraction from healthcare network traffic.

focus of this research is the intermittent connectivity of clients to a variety of cybersecurity organizations. Furthermore, for local training, each client may have access to imbalanced malware datasets. We developed a deep CNN-based FL approach to address the challenges of intermittent clients and imbalanced datasets.

III. PROPOSED METHOD: FEDERATED EDGE INTELLIGENCE FOR HEALTHCARE SECURITY

A. Visual Features Extraction From Network Patterns

Figure 2 shows the architectural framework of the proposed approach.

PCAP files are critical to improving network security and speed. They give detailed logs of network packets, which aid in detecting irregularities and threats. These files are useful for investigating events by tracing breaches or network difficulties. They also support protocol analysis, which aids in the detection of errors and unauthorized use. Performance evaluation is enhanced by analyzing packet data to discover bottlenecks and latency concerns. Additionally, PCAP files provide an audit record for regulatory compliance and security policy enforcement. They help with threat detection by highlighting unusual behaviors and significant security threats. Moreover, PCAP is useful for debugging network issues because it provides specific data flows. Considering the continual upgrading of network traffic to avoid static and dynamic classification, we investigated the usefulness of visual features in detecting malicious traffic. This method uses image processing to transform network traffic patterns into textural properties. It works without involving reverse engineering or traffic signatures. This solution successfully reduces the impact of anti-detection strategies including signature modification and dynamic feature extraction evasion [19]. By examining the PCAP, a byte stream about each malicious variation is obtained. We designed a bytes-to-image translation technique for extracting images from byte streams. Greyscale images are produced by processing 8-bit vectors retrieved from network-based byte streams. Following that, each image is set to 256x256 pixels. Figure 3 depicts an example of 256x256 network traffic images for spoofing, Distributed Denial of Service (DDoS), Message Queuing Telemetry Transport (MQTT), and DoS. Smaller image sizes have been found to result in a significant reduction in network traffic. For instance, the picture converts megabytes of PCAP data into kilobytes. This has the potential to reduce computational power.

B. Data Augmentation

Data augmentation improves model performance and addresses data imbalance in real-world datasets, especially

Algorithm 1: Generating Images From Network Traffic

Input: F_{PCAP} : Folder containing PCAP files

Output: Images representing byte streams

Function GenerateImages (F_{PCAP}):

```

for each PCAP file  $pcap\_file \in F_{PCAP}$  do
    Open  $pcap\_file$ ;
    Initialize  $accumulated\_bytes \leftarrow \{\}$ ;
    Initialize  $total\_bytes \leftarrow 65536$ ;
    Initialize  $packet\_count \leftarrow 0$ ;
    for each packet  $p_i$  in  $pcap\_file$  do
        byte stream:  $bytestream_i$ ;
        Accumulate byte streams
         $\sum_{j=1}^n |bytestream_j| \geq total\_bytes$ ;
        if
             $\sum_{j=1}^k |accumulated\_bytes_j| + |bytestream_i| \geq total\_bytes$  then
                Append  $bytestream_i$  to
                     $accumulated\_bytes$ ;
                 $remaining\_bytes \leftarrow$ 
                     $total\_bytes - \sum_{j=1}^k |accumulated\_bytes_j|$ ;
                Truncate  $bytestream_i$  to fit remaining
                    space;
                Create image  $accumulated\_bytes$ ;
                Save image;
                Increment  $packet\_count$ ;
                Clear  $accumulated\_bytes$ ;
                Append  $bytestream_i$  to
                     $accumulated\_bytes$ ;
    Save images to corresponding directories;

```

when identifying adverse network traffic on healthcare equipment [20]. It involves transforming and modifying the training dataset $\mathcal{D}$ to increase diversity and allow machine learning models to learn from diverse network traffic patterns. To illustrate how data transmission and packet structure can vary, alternative orientations and sizes of network traffic images can be simulated using random rotations (θ), flips, and scaling ($resize(x, y)$). The following mathematical expressions describe these transformations using equations (1)-(2). - Rotating an image I by angle θ :

$$I_{rotated} = rotate(I, \theta). \quad (1)$$

- Resizing an image I to dimensions $x \times y$:

$$I_{resized} = resize(I, x, y). \quad (2)$$

Additionally, adding random noise (ϵ) with mean μ and standard deviation σ improves the model's capacity to detect fragile network traffic anomalies. This noise enhancement is especially useful for medical devices since it mimics actual changes and makes the model more resistant to chaotic network conditions. Adding noise to an image I can be mathematically expressed using equation (3).

$$I_{noisy} = I + \epsilon, \quad \epsilon \sim \mathcal{N}(\mu, \sigma). \quad (3)$$

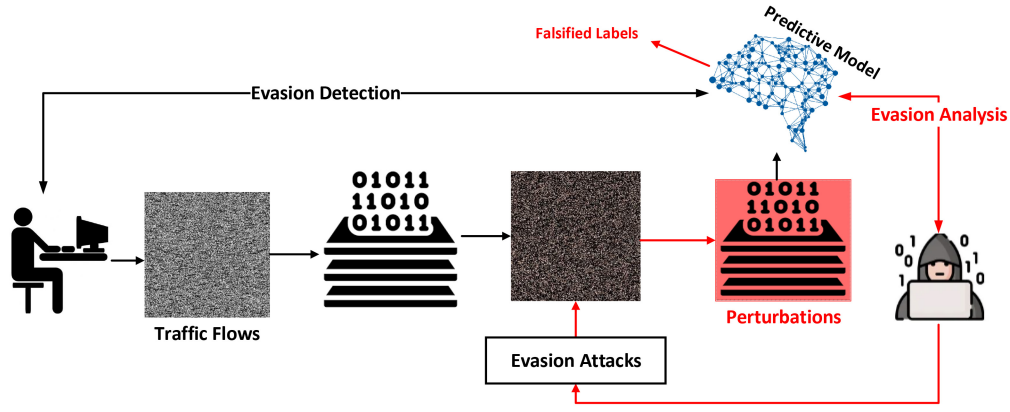


Fig. 4. Overview of evasion attacks using abnormal network traffic flows.

Additionally, the dataset is made more diverse by methods like color jittering and cropping ($\text{crop}(x, y)$), which help the model acquire invariant properties across various spatial layouts and color distributions. The following can be used to crop an image I to dimensions $x \times y$ using equation (4).

$$I_{\text{cropped}} = \text{crop}(I, x, y). \quad (4)$$

Color jittering, which alters brightness, contrast, saturation, and hue, can be represented using equation (5).

$$I_{\text{jittered}} = \text{jitter}(I, \delta_{\text{brightness}}, \delta_{\text{contrast}}, \delta_{\text{saturation}}, \delta_{\text{hue}}). \quad (5)$$

Data augmentation enhances machine learning models by adding diverse transformations, helping them learn robust features for better and more reliable detection of malicious activity in healthcare devices [21]. To boost the performance of our malware detection model, we applied data augmentation techniques such as rotation, jitter, noise addition, cropping, and resizing. These strategies are critical in increasing the diversity and variability of the training dataset, improving its robustness and generalization. By simulating numerous real-world characteristics such as evolving image orientation, position, and noise levels, these augmentations enabled our model to handle a broader range of scenarios and reduce overfitting. This method is particularly helpful given the limited data available and the necessity for the model to work effectively in changing scenarios.

C. Crafting Adversarial Examples

We used and evaluated four adversarial attack techniques: PGD, FGSM, IGSM, and a novel hybrid strategy that combined PGD and IGSM [22], [23]. As shown in Figure 4, we aim to assess how well these strategies generate adversarial examples that mislead a target model with few perturbations.

1) *Fast Gradient Sign Method (FGSM)*: The FGSM algorithm perturbs input data $\mathbf{x}$ in the direction of the gradient of the loss function $J(\theta, \mathbf{x}, y)$. It is simple and computationally efficient. The strength of the threat is regulated by the variable ϵ , which scales the perturbation. The following is the process for generating the adversarial example, $\mathbf{x}'$ using equation (6).

$$\mathbf{x}' = \mathbf{x} + \epsilon \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}, y)). \quad (6)$$

The one-step simplicity of FGSM makes it excellent at producing adversarial samples rapidly, but it is typically ineffective against well-defended systems.

2) *Iterative Gradient Sign Method (IGSM)*: IGSM, which stands for Basic Iterative Method (BIM), is an extension of FGSM that uses iterative applications to achieve a cumulative effect with fewer perturbations at each stage. The following is how the technique improves the adverse effects after several iterations k using equations (7) and (8).

$$\mathbf{x}'^{(0)} = \mathbf{x}. \quad (7)$$

$$\mathbf{x}'^{(k+1)} = \text{clip}_{\mathbf{x}, \epsilon} \left\{ \mathbf{x}'^{(k)} + \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}'^{(k)}, y)) \right\}. \quad (8)$$

The step size is denoted by α , and perturbations are guaranteed to be kept within a ϵ -ball surrounding the initial input $\mathbf{x}$.

3) *Projected Gradient Descent (PGD)*: PGD is an iterative approach that functions as a resilient complement to IGSM. PGD employs FGSM at each iteration, then performs an estimating step to guarantee perturbations stay within a defined ℓ_p norm ball. Here is the sequence of iterations using equations (9) and (10)

$$\mathbf{x}'^{(0)} = \mathbf{x}. \quad (9)$$

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathbf{x} + \mathcal{B}_p(\epsilon)} \left(\mathbf{x}'^{(k)} + \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}'^{(k)}, y)) \right). \quad (10)$$

The projection onto the norm ball of radius ϵ around $\mathbf{x}$ is represented by $\Pi_{\mathbf{x} + \mathcal{B}_p(\epsilon)}$. It confirms that the adversarial instances work as expected and remain within a reasonable perturbation bound.

4) *Hybrid Approach of PGD and IGSM*: The hybrid methodology leverages the distinctive features of both PGD and IGSM. We utilize IGSM to generate adversarial instances and initiate the attack in this method. Then, it is imperative to optimize these examples further using PGD to ensure that they are within the intended perturbation limits and to enhance their efficacy. The process can be described as follows using equations (11), (12) and (13).

- IGSM Initialization:

$$\mathbf{x}'^{(0)} = \mathbf{x}. \quad (11)$$

$$\mathbf{x}'^{(k+1)} = \text{clip}_{\mathbf{x}, \epsilon} \left\{ \mathbf{x}'^{(k)} + \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}'^{(k)}, y)) \right\}. \quad (12)$$

- PGD Refinement:

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathbf{x}+\mathcal{B}_p(\epsilon)}\left(\mathbf{x}'^{(k)} + \alpha \cdot \text{sign}\left(\nabla_{\mathbf{x}}J(\theta, \mathbf{x}'^{(k)}, y)\right)\right). \quad (13)$$

This hybrid strategy combines the robust estimation of PGD with the continuous refining of IGSM, leveraging the strengths of both techniques [24], [25]. The adversarial attack generation technique that combines PGD and IGSM surpasses distinct methods such as FGSM, IGSM, and PGD. The ability of PGD to undertake significant gradient-based optimization, along with the iterative improvement of IGSM, improves the precision and effectiveness of adversarial instances. PGD provides a strong starting estimation, and IGSM refines these estimates to improve robustness and accuracy. Performance is greatly enhanced by this synergy, which guarantees that the adversarial examples generated are both extremely effective and resilient within the given limitations.

D. Federated-Based Edge Computing: Adversarial Attacks Detection and Classification

FL provides edge intelligence by processing data locally on each client device, reducing latency and enhancing privacy. This is particularly helpful as it allows for real-time detection and classification of adversarial attacks directly on healthcare devices without sending sensitive data to a central server [26]. This research aims to leverage secure FL settings for detecting and classifying adversarial attacks on healthcare devices within the context of cybersecurity. McMahan et al. [27] introduced the FL approach for data-driven system design using distributed data without relying on a centralized data center. This collaborative learning framework, which requires only the sharing of model updates, can significantly enhance security and privacy, particularly for healthcare devices more susceptible to cyberattacks. A timely and comprehensive understanding of cyberattacks, such as spoofing, intrusion, anomaly, and Denial of Service (DoS), is crucial for developing and improving cyber defense models and methods. Consequently, FL has tremendous potential to secure cyberspace at device and network levels.

- 1) Localized Training Process: In a FL scenario, a soft federation of distinct entities or machines, referred to as clients, is coordinated to train a model. In our case, these clients could be various medical institutions or devices using adversarial network images generated on medical devices [28]. Each client maintains local training data, known as Local Model Update (LMU), which updates the current global model. Therefore, client data is never transmitted to the global server; only the latest model update is shared. Mathematically, the update process for the global model w at time t can be expressed using equation (14)

$$w_{t+1} = w_t - \eta \cdot \nabla f(w_t). \quad (14)$$

where η is the learning rate, $\nabla f(w_t)$ is the gradient of the loss function concerning the model parameters w , computed based on the local data of each client. The healthcare devices, acting as clients, conduct mini-batch Stochastic Gradient Descent (SGD) on locally collected images using deep CNN. Our method employs sporadic

clients, represented by healthcare devices, sharing the same CNN framework and loss functions. Each device retrieves global model weights and computes a gradient update after multiple SGD iterations, ensuring data confidentiality. CNN, widely used for detection tasks, extracts information for deep training and potential gains in model capacity. Our deep CNN framework processes images on healthcare devices to produce LMU. It includes three convolutional layers with filter sizes of 64, 128, 256, and 512, respectively, followed by pooling layers, dropout layers, a fully connected layer, and a softmax layer. Batch normalization ensures training stability, reducing overfitting risk [29], [30].

- 2) Global Training Process: Several causes can contribute to intermittent behavior among remote customers in FL, including dropped connections, delayed data delivery, and system failures. Existing methods often fail to account for this intermittency, resulting in a loss in model accuracy. To solve this issue, we present a FL system capable of detecting adversarial attacks on a variety of healthcare devices. It is critical to properly manage clients' inconsistent behavior, including their decision to engage in or withdraw from training. The inclusion of a new client and the removal of an existing one may have an impact on the dataset balance. To accommodate irregular clients and unbalanced datasets, we suggest storing and using weights from paused clients for further aggregations, or ignoring weights from departed clients in subsequent iterations. The best course of action is context-dependent; variables such as dataset imbalance and client engagement influence the relative advantages of the two strategies.

In FL, the global server handles dynamic client behavior caused by network issues and delays by employing asynchronous aggregation, which allows it to incorporate updates from clients at various times. It properly schedules clients, uses robust aggregation methods to handle delays and lost updates, and adjusts learning rates based on client behavior. Furthermore, despite fluctuations in client connectivity and availability, the training process is made durable and successful by employing appropriate communication protocols and fault recovery methods. To assess our method, we conducted extensive experiments with intermittent clients and varied adversarial attack image sizes. These studies give information on the efficacy of various ways for minimizing the effects of intermittent client behavior on FL systems. The heterogeneity in network data samples often leads to a lack of generalization. Consequently, a model with high performance in one network may be inefficient at detecting attacks in another. This is due to each organizational network having its own Standard Operating Environments (SOEs) [31] and various potential attacks. The conditional distribution of the used data reflects these threats. Machine learning models heavily rely on mining useful features to differentiate between benign and malicious samples. Therefore, training a threat detection model necessitates a broad range of data samples. In ML-based attack detection, data scarcity and heterogeneity are critical considerations in each ML scenario [32], [33].

IV. RESULTS AND DISCUSSIONS

A. Dataset Preparation

We evaluated the proposed approach with two standard and realistic datasets, such as CICIoMT2024¹ [34], and Edge-IIoT² [35]. The CICIoMT2024 dataset includes healthcare equipment attack vectors. The dataset evaluates IoMT security by analyzing attacks on 40 devices (25 real and 15 simulated) using healthcare protocols such as Wi-Fi, Queuing Telemetry Transport (MQTT), and Bluetooth. There have been five identified attacks: DDoS, DoS, recon, MQTT, and spoofing. A network tap created a security and profiling dataset by replaying real-time packets between the switch and Wi-Fi/MQTT-capable IoMT devices. Edge-IIoT, the second dataset, was developed using an IoT/IIoT testbed, which included devices, sensors, protocols, and edge configurations. This compilation comprises data from over 10 IoT devices, including low-cost digital temperature and humidity sensors. There are 14 attacks on IoT and IIoT communication protocols, including DoS/DDoS, information gathering, man-in-the-middle, Injection, and Malware. Alarms, system resources, logs, and network traffic datasets have 61 significantly related features from 1176.

B. Performance Indicators

We selected various network traffic-based images and several clients to evaluate the proposed method comprehensively. The proposed method is evaluated using two publicly available standard datasets, CICIoMT2024 and Edge-IIoT. We measured performance using precision, recall, F-measure, Kappa, and accuracy. These metrics are calculated from True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) counts. The performance measures are provided by Equations (15)-(19).

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (15)$$

$$\text{Recall} = \frac{FP}{(FP + TN)} \quad (16)$$

$$\text{F-measure} = \frac{(2 * TP)}{(2TP + FP + FN)} \quad (17)$$

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (18)$$

$$\text{Kappa} = \frac{(\text{observed}_{acc} - \text{expected}_{acc})}{(1 - \text{expected}_{acc})} \quad (19)$$

where, $\text{total} = TP + TN + FP + FN$, $\text{observed}_{acc} = (TP + TN)/\text{total}$, $a = (TP + FN) * (TP + FP)/\text{total}$, $b = (FP + TN) * (FN + TN)/\text{total}$, and $\text{expected}_{acc} = (a + b)/\text{total}$.

C. Performance Analysis and Comparisons

The proposed method is tested on various clients to examine intermittent behavior. Figures 5-8 illustrate the local and global accuracy curves used to study the dynamic behavior

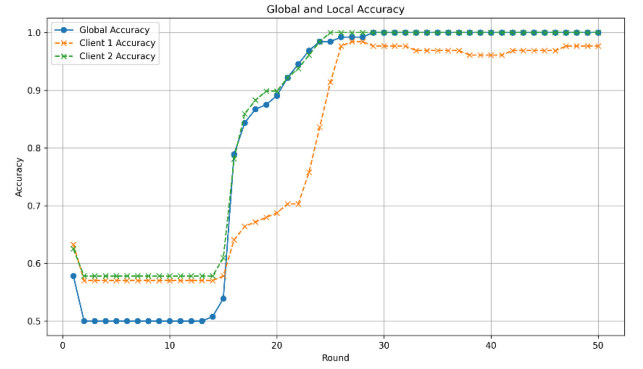


Fig. 5. Dynamic local and global accuracy curves using 2 clients.

of the proposed system. The proposed method is tested with clients numbered 2, 3, 10, and 15. It is determined that expanding the number of clients yields the best improvements. Having more FL clients boosts performance due to the diversity of datasets, parallel processing, integrated knowledge, confidentiality, and reliability. Figure 5 illustrates the two clients' local and global accuracy curves. The green and orange colors represent the clients, while the blue represents the global averaged accuracy values. Both clients start at around 63% accuracy, but the global server starts at 58%. After a few epochs, client 1 has worse performance than client 2. Client 1 has 70% accuracy, whereas client 2 has 92% on the 22nd epoch. Furthermore, the global accuracy curve performs similarly depending on client 2. Overall, client 1 performs poorly because it frequently loses precision. Figure 6 illustrates three clients' local and global accuracy values. When compared to each other, their dynamic values vary greatly. Client 1 has the lowest performance, client 3 has the highest, and client 2 is somewhat in the middle. The global accuracy curve also performs roughly in the middle. For instance, on epoch 20, client 1 has 65% accuracy, client 3 has 99%, client 2 has 87%, and global accuracy is 77%. After the 30th epoch, the three clients and the global server are behaving similarly. Figure 7 depicts the local and global accuracy curves for ten clients. It demonstrates significant variances in client performance, although the global server performs better after the 23rd epoch. Our key goal is for the global server to perform better by averaging client weights and leveraging varied datasets. Similarly, 8 illustrates the dynamic local and global accuracy curves with 15 clients. Overall, the global server improves after the 28th epoch.

Figures 9-12 depict the dynamic local and global loss values. Figure 9 analyzes two clients with a global server. The major purpose is to obtain the declining trend for each epoch. For instance, client 1, client 2, and the global server begin with a 70% loss and steadily reduce. Client 1 displays slightly larger loss values than client 2 and the global server. At the 38th epoch, client 1 has a 10% loss, whereas client 2 has 2%. Figure 10 illustrates three clients' local and global loss values. Client 1 has the biggest loss of over 100% on the 24th epoch, while client 3 has the lowest loss of 2%. Client 2 and the global server both show slightly higher second-best losses. Figures 11 and 12 demonstrate loss values for ten and fifteen clients, respectively. It can be observed that the global server

¹<https://www.unb.ca/cic/datasets/iomt-dataset-2024.html>

²<https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iiot-iiot/code>

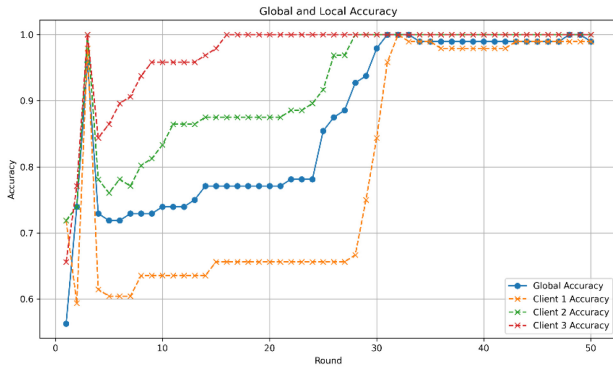


Fig. 6. Dynamic local and global accuracy curves using 3 clients.

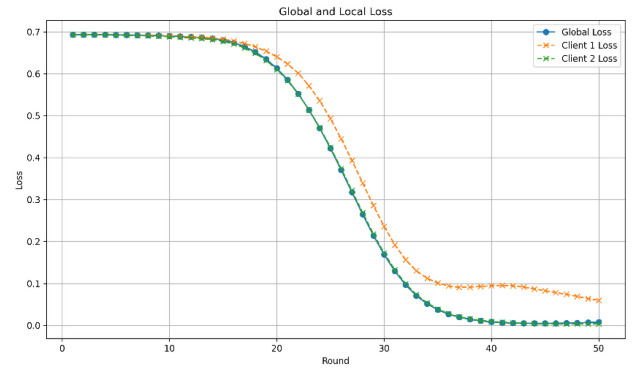


Fig. 9. Dynamic local and global loss curves using 2 clients.

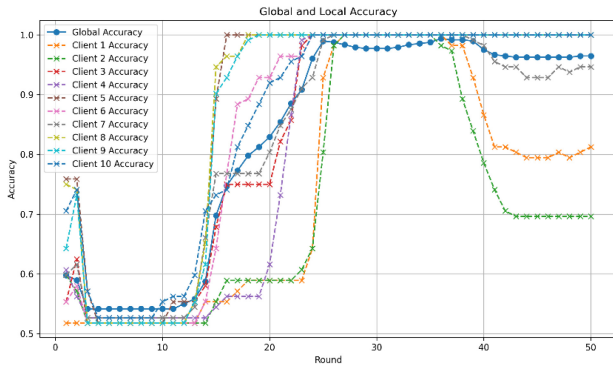


Fig. 7. Dynamic local and global accuracy curves using 10 clients.

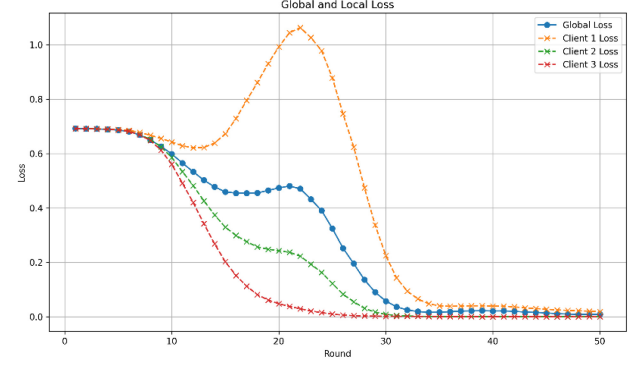


Fig. 10. Dynamic local and global loss curves using 3 clients.

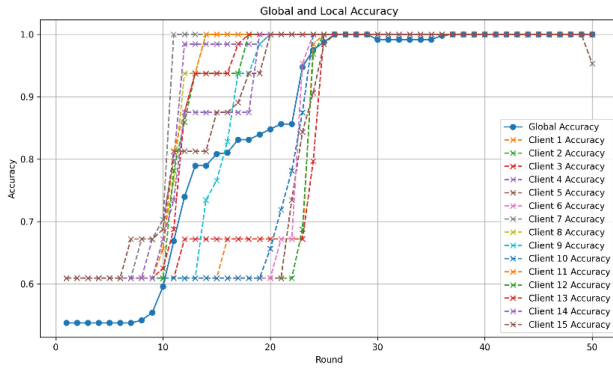


Fig. 8. Dynamic local and global accuracy curves using 15 clients.

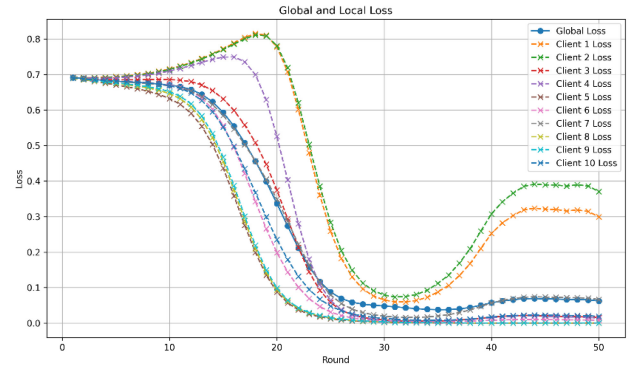


Fig. 11. Dynamic local and global loss curves using 10 clients.

always performs better with a larger number of clients after a few epochs, which is the primary purpose of the proposed approach. Table I compares the performance of several forms of adversarial attack detection on the CICIOMT2024 dataset. The FGSM attack is identified with reduced performance, whereas the hybrid is detected with maximum performance. Although the hybrid attack is the most powerful feature attack due to the combination of IGSM and PGD. The proposed method remains the most effective. For instance, the FGSM attack has precision, recall, and f1-score, with accuracy of 93.2%, 97.2%, 95.6%, and 95.2%. The performance measures for hybrid attacks are 98.3%, 100%, 99%, and 99.34%. The PGD attack is detected more than the other two approaches (FGSM and IGSM), but less than the hybrid strategy. For

TABLE I
PERFORMANCE COMPARISONS FOR ADVERSARIAL ATTACKS
DETECTION USING CICIOMT2024

Adversarial	Traffic	Precision	Recall	F1 Score	Accuracy
FGSM	Benign	96.0	94.3	95.0	95.2
	Attack	93.2	97.2	95.6	
IGSM	Benign	99.0	95.0	97.2	96.73
	Attack	95.7	99.1	97.5	
PGD	Benign	100	97.4	99.3	98.86
	Attack	98.4	100	99.0	
Hybrid	Benign	100	98.0	99.2	99.34
	Attack	98.3	100	99.0	

instance, the PGD has precision, recall, f1-score, and accuracy of 98.3%, 100%, 99%, and 99.34%, respectively.

Table II compares the performance of several attack detection methods using the Edge-IIoT dataset. Similar to Table I,

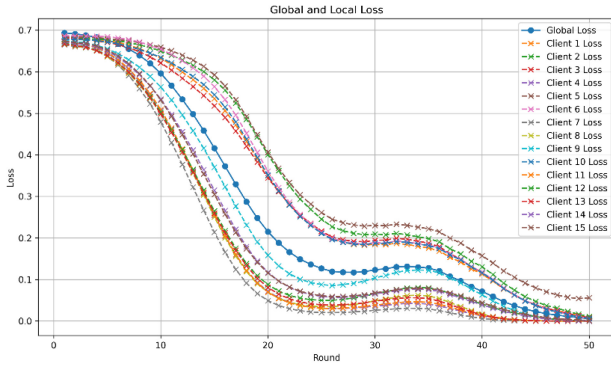


Fig. 12. Dynamic local and global loss curves using 15 clients.

TABLE II
PERFORMANCE COMPARISONS FOR ADVERSARIAL ATTACKS DETECTION
USING EDGE-IIoTSET

Adversarial	Traffic	Precision	Recall	F1 Score	Accuracy
FGSM	Benign	94.8	98.3	96.0	95.2
	Attack	97.0	94.7	96.1	
IGSM	Benign	93.9	100	97.0	96.1
	Attack	100	93.4	95.8	
PGD	Benign	98.8	99.0	99.0	99.1
	Attack	99.2	99.4	98.7	
Hybrid	Benign	100	99.0	99.6	99.2
	Attack	98.9	100	98.8	

hybrid attacks are identified more effectively than other methods. For the hybrid attack, the precision, recall, f1 score, and accuracy are 98.9%, 100%, 98.8%, and 99.2%. Similarly, the same performance measurements are lowest for FGSM, but PGD is superior. Table III presents performance metrics, including precision, recall, and f1 score for classifying adversarial attacks on the CICIOMT2024 dataset. There are five classes: DDoS, DoS, MQTT, Recon, and Spoofing. As can be observed, hybrid adversarial attacks are accurately divided into five distinct attacks. For instance, DDoS (94.7%, 96%, 96.1%), DoS (97.6%, 98.1%, 97.6%), MQTT (95.8%, 98.7%, 98.3%), Recon (98.2%, 94%, 95.9%), and Spoofing (100%, 100%, 100%). Table IV shows the performance classification of several adversarial attacks on the Edge-IIoT dataset. We presented adversarial examples for FGSM and hybrid to demonstrate the effectiveness of the proposed approach. The Edge-IIoT dataset has 14 categories, and the suggested strategy is more accurate in classifying hybrid-based adversarial attacks. Table V presents classification accuracy utilizing four adversarial attacks for the suggested technique. The hybrid technique achieves the greatest classification rate of 97.45% utilizing the CICIOMT2024 dataset and 96.7% with the Edge-IIoTset dataset.

To provide deeper analysis, the confusion matrix can show the correct and incorrect classification rates for each class in the dataset. Figure 13 presents the confusion matrix for FGSM-based adversarial attack detection with the CICIOMT2024 dataset. The benign and attack are successfully identified at 92% and 97%, respectively. Figure 14 depicts the confusion matrix for hybrid-based adversarial attack detection using the CICIOMT2024 dataset. The benign and attack are appropriately identified at 98% and 100% respectively. Figure 15 depicts the confusion matrix for the hybrid adversarial approach with the Edge-IIoTset dataset. Figure 16 shows

TABLE III
PERFORMANCE COMPARISONS FOR ADVERSARIAL ATTACKS
CLASSIFICATION USING CICIOMT2024

Adversarial	Attack	Precision	Recall	F1 Score
FGSM	DDoS	79.9	92.1	85.8
	DoS	92.1	83.4	87.6
	MQTT	87.8	91.0	89.3
	Recon	99.0	92.7	94.8
	Spoofing	100	100	100
IGSM	DDoS	85.0	91.1	87.0
	DoS	89.8	88.3	88.9
	MQTT	93.5	95.0	94.0
	Recon	99.0	91.7	95.1
	Spoofing	100	100	100
PGD	DDoS	93.1	88.7	91.1
	DoS	96.8	93.2	95.0
	MQTT	94.5	98.4	97.6
	Recon	91.0	94.4	93.7
	Spoofing	100	100	100
Hybrid	DDoS	94.7	96.0	96.1
	DoS	97.6	98.1	97.6
	MQTT	95.8	98.7	98.3
	Recon	98.2	94.0	95.9
	Spoofing	100	100	100

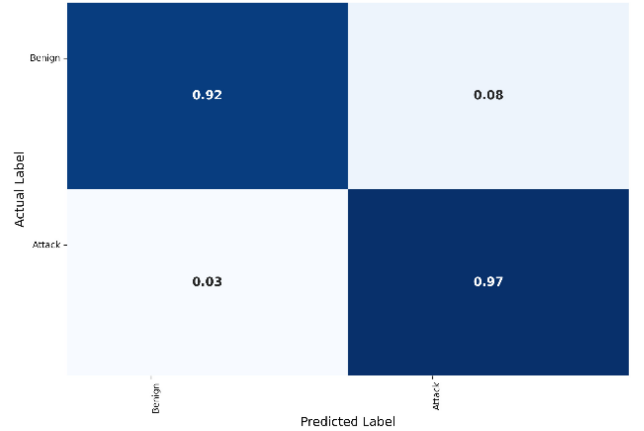


Fig. 13. FGSM-based Adversarial attacks detection using CICIOMT2024 dataset.

the confusion matrix for FGSM-based adversarial attack classification on the CICIOMT2024 dataset. DDoS, DoS, MQTT, Recon, and Spoofing are correctly characterized at 92%, 83%, 91%, 92%, and 100%. Figure 17 illustrates the confusion matrix for hybrid adversarial attack classification. DDoS, DoS, MQTT, Recon, and Spoofing are accurately identified at 96%, 98%, 99%, 94%, and 100%. Figure 18 presents the confusion matrix for adversarial attack classification using the Edge-IIoTset dataset.

Table VI presents a performance comparison with related studies. Fan et al. [9] introduced a method for generating adaptable adversarial instances without a replacement model, achieving a 95.16% success rate. Their approach involves image-to-graph transformation, perturbation on the graph, and graph-to-image inversion. Guduri et al. [14] developed FedKC, a personalized FL algorithm for resisting model poisoning in the consumer health metaverse. FedKC provides tailored medical treatments for new users and demonstrated 91.1% accuracy in mitigating attacks and enhancing client customization in experimental tests. Mothukuri et al. [18] developed a method for detecting FL anomalies in IoT networks using GRUs and decentralized on-device data, sending only learned

TABLE IV
PERFORMANCE COMPARISONS FOR ADVERSARIAL ATTACKS CLASSIFICATION USING EDGE-IIoTSET

Adversarial	Attack	Precision	Recall	F1-score
FGSM	Backdoor	68.8	74.1	71.0
	DDoS HTTP Flood	83.4	91.2	87.5
	DDoS ICMP Flood	89.0	96.1	92.1
	DDoS TCP SYN Flood	94.8	86.3	91.0
	DDoS UDP Flood	98.0	99.0	99.1
	MITM (ARP spoofing + DNS)	78.0	88.9	83.5
	OS Fingerprinting	96.6	99.4	98.3
	Password	90.7	57.4	70.1
	Port Scanning	99.1	95.4	97.1
	Ransomware	98.0	100	98.9
	SQL injection	75.4	78.2	77.5
	Uploading	91.3	90.2	90.0
	Vulnerability scanner	0.99	99.3	998.7
	XSS	0.99.0	98.4	98.8
Hybrid	Backdoor	89.1	96.4	91.6
	DDoS HTTP Flood	97.4	97.1	96.6
	DDoS ICMP Flood	98.1	97.6	98.0
	DDoS TCP SYN Flood	91.7	94.4	93.1
	DDoS UDP Flood	99.8	100	100
	MITM (ARP spoofing + DNS)	84.5	94.8	89.0
	OS Fingerprinting	98.2	100	99.6
	Password	96.1	80.6	0.88
	Port Scanning	100	95.3	96.9
	Ransomware	0.98	100	0.99
	SQL injection	100	95.7	98.3
	Uploading	98.4	99.0	99.0
	Vulnerability scanner	100	99.1	98.6
	XSS	99.3	100	99.0

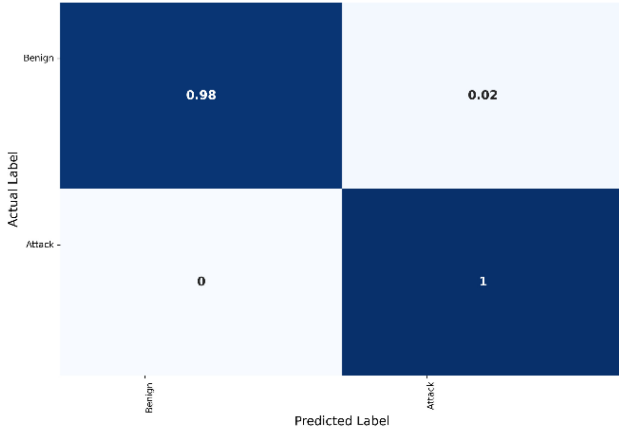


Fig. 14. Hybrid-based Adversarial attacks detection using CICIoMT2024 dataset.

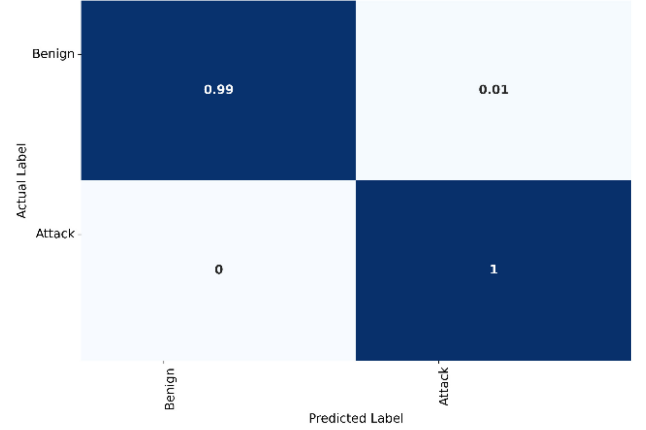


Fig. 15. Hybrid-based Adversarial attacks detection using Edge-IIoTset dataset.

weights to the FL server. This approach improves global ML model accuracy and offers superior privacy protection and attack detection, achieving 95.6% accuracy. Zhang et al. [36] used GANs to study poisoning threats in FL, where attackers simulate participants to generate poisoned updates from copied samples. Their method can produce samples with over 80% accuracy for both poisoning and main activities. The proposed approach provides better performance, i.e., 97.45% as compared to the related works.

V. CONCLUSION

SHS is a contemporary consumer electronics concept that emerged from the integration of networked personal medical devices. These systems collected data, sent it to smart devices for analysis, and delivered healthcare services using edge

TABLE V
CLASSIFICATION ACCURACY USING CICIoMT2024 AND EDGE-IIoTSET

Datasets	Adversarial Method	Accuracy
CICIoMT2024	FGSM	92.3
	IGSM	93.11
	PGD	95.18
	Hybrid	97.45
Edge-IIoTset	FGSM	90.14
	IGSM	91.33
	PGD	94.52
	Hybrid	96.7

computing and medical equipment. However, the possibility of cyberattacks increased as a result of this interconnectedness, especially in vital industries like healthcare, where attacks on consumer medical equipment were common. Massive volumes of data from medical devices also offered significant

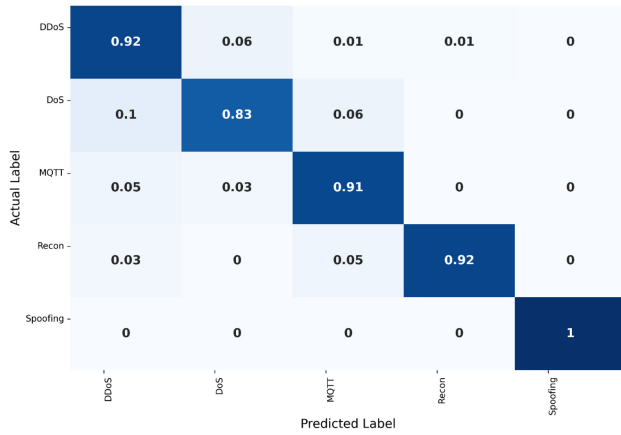


Fig. 16. FGSM-based Adversarial attacks classification using CICIoMT2024 dataset.

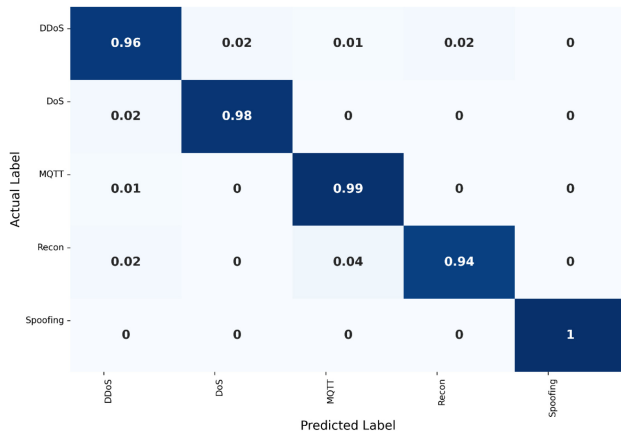


Fig. 17. Hybrid-based Adversarial attacks classification using CICIoMT2024 dataset.

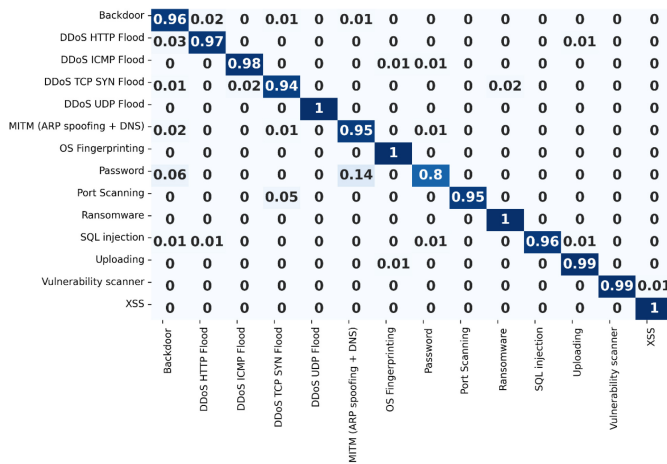


Fig. 18. Hybrid-based Adversarial attacks classification using Edge-IIoTset dataset.

challenges in terms of storage and security. Utilizing edge computing as a layer between the cloud and SHS is crucial to overcoming these challenges. The proposed method enhanced security and privacy while simplifying the processing of large volumes of data. We used an edge intelligence approach that included suspicious examples of network traffic and FL to

TABLE VI
PERFORMANCE COMPARISON WITH RECENT AND RELATED WORKS

Work	Method	Performance
Fan et al. [10]	Graph-based Sampling	95.16
Le et al. [13,2]	Personalized FL	91.4
Viraaji et al. [17]	GRUs with FL	95.65
Zhang et al. [34]	GANs-based FL	80.0
Our Method	Adversarial examples with FL	97.45

solve these issues and enhance the security of medical devices in SHS. On standard datasets, CICIoMT2024 and Edge-IIoT detect and mitigate adversarial attacks with 97.45% and 96.7% accuracy, respectively. This emphasizes the crucial need for creative solutions that improve healthcare data security while assuring the continued delivery of essential healthcare services. Our work protects sensitive healthcare data and strengthens SHSs against cyber-attacks, hence improving the security of existing healthcare technology. FL protects medical data by storing it on local devices and delivering only model updates instead of raw data. Security and privacy can be enhanced by the use of blockchain technology, homomorphic encryption, safe aggregation, and differential privacy. While homomorphic encryption allows computations on encrypted data, secure aggregation uses cryptographic algorithms to protect updates, and blockchain ensures apparent process recording, differential privacy adds noise to model updates. Data privacy is further enhanced by regular security audits and regulatory compliance.

REFERENCES

- [1] A. Rahman, M. S. Hossain, N. A. Alrajeh, and F. Alsolami, "Adversarial examples—Security threats to COVID-19 deep learning systems in medical IoT devices," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9603–9610, Jun. 2021.
- [2] H. R. Chi, M. de Fátima Domingues, H. Zhu, C. Li, K. Kojima, and A. Radwan, "Healthcare 5.0: In the perspective of consumer Internet-of-Things-based fog/cloud computing," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 745–755, Nov. 2023.
- [3] T. Yaqoob, H. Abbas, and M. Atiquzzaman, "Security vulnerabilities, attacks, countermeasures, and regulations of networked medical devices—A review," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3723–3768, 4th Quart., 2019.
- [4] D. Li, Q. Li, Y. Ye, and S. Xu, "A framework for enhancing deep neural networks against adversarial malware," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 1, pp. 736–750, Jan.–Mar. 2021.
- [5] L. Lyu et al., "Privacy and robustness in federated learning: Attacks and defenses," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 8726–8746, Jul. 2024.
- [6] F. Ullah, G. Srivastava, S. Ullah, and L. Mostarda, "Privacy-preserving federated learning approach for distributed Malware attacks with intermittent clients and image representation," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 4585–4596, Feb. 2024.
- [7] A. I. Newaz, A. K. Sikder, M. A. Rahman, and A. S. Uluagac, "A survey on security and privacy issues in modern healthcare systems: Attacks and defenses," *ACM Trans. Comput. Healthc.*, vol. 2, no. 3, pp. 1–44, 2021.
- [8] A. I. Newaz, A. K. Sikder, L. Babun, and A. S. Uluagac, "HEKA: A novel intrusion detection system for attacks to personal medical devices," in *Proc. IEEE Conf. Commun. Netw. Security (CNS)*, 2020, pp. 1–9.
- [9] X. Fan et al., "GCSA: A new adversarial example-generating scheme toward black-box adversarial attacks," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2038–2048, Feb. 2024.
- [10] M. Zhang, Y. Duan, H. Yin, and Z. Zhao, "Semantics-aware android malware classification using weighted contextual API dependency graphs," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2014, pp. 1105–1116.

- [11] H. Gao, S. Cheng, and W. J. C. Zhang, "GDroid: Android malware detection and classification with graph convolutional network," *Comput. Secur.*, vol. 106, Jul. 2021, Art. no. 102264.
- [12] X. Zhang, L. Hao, G. Gui, Y. Wang, B. Adebisi, and H. Sari, "An automatic and efficient malware traffic classification method for secure Internet of Things," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 8448–8458, Mar. 2024.
- [13] R. Kumari, D. K. Sah, S. Gupta, K. Cengiz, and N. Ivković, "Advancing medical recommendations with federated learning on Decentralized data: A roadmap for implementation," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2666–2674, Feb. 2024.
- [14] M. Guduri, C. Chakraborty, U. Maheswari, and M. Margala, "Blockchain-based federated learning technique for privacy preservation and security of smart electronic health records," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2608–2617, Feb. 2024.
- [15] W. Schneble and G. Thamilarasu, "Attack detection using federated learning in medical cyber-physical systems," in *Proc. 28th Int. Conf. Comput. Commun. Netw. (ICCCN)*, vol. 29, 2019, pp. 1–8.
- [16] A. Rehman, I. Razzak, and G. Xu, "Federated learning for privacy preservation of healthcare data from smartphone-based side-channel attacks," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 2, pp. 684–690, Feb. 2023.
- [17] W. Wang, X. Li, X. Qiu, X. Zhang, V. Brusica, and J. Zhao, "A privacy preserving framework for federated learning in smart healthcare systems," *Inf. Process. Manage.*, vol. 60, no. 1, 2023, Art. no. 103167.
- [18] V. Mothukuri, P. Khare, R. M. Parizi, S. Pouriyeh, A. Dehghantanha, and G. Srivastava, "Federated-learning-based anomaly detection for IoT security attacks," *IEEE Internet Things J.*, vol. 9, no. 4, pp. 2545–2554, Feb. 2022.
- [19] S. Ni, Q. Qian, and R. Zhang, "Malware identification using visualization images and deep learning," *Comput. Secur.*, vol. 77, pp. 871–885, Aug. 2018.
- [20] P. Chlap, H. Min, N. Vandenberg, J. Dowling, L. Holloway, and A. Haworth, "A review of medical image data augmentation techniques for deep learning applications," *J. Med. Imag. Radiat. Oncol.*, vol. 65, no. 5, pp. 545–563, 2021.
- [21] E. Lashgari, D. Liang, and U. Maoz, "Data augmentation for deep-learning-based electroencephalography," *J. Neurosci. Methods*, vol. 346, Dec. 2020, Art. no. 108885.
- [22] A. Mustafa, S. Khan, M. Hayat, R. Goecke, J. Shen, and L. Shao, "Adversarial defense by restricting the hidden space of deep neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 3385–3394.
- [23] Y. Shi, Y. Han, Q. Hu, Y. Yang, and Q. Tian, "Query-efficient black-box adversarial attack with customized iteration and sampling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2226–2245, Feb. 2023.
- [24] Y. Zhao, K. Xu, H. Wang, B. Li, M. Qiao, and H. Shi, "MEC-enabled hierarchical emotion recognition and perturbation-aware defense in smart cities," *IEEE Internet Things J.*, vol. 8, no. 23, pp. 16933–16945, Dec. 2021.
- [25] F. Chen et al., "Frequency constraint-based adversarial attack on deep neural networks for medical image classification," *Comput. Biol. Med.*, vol. 164, Sep. 2023, Art. no. 107248.
- [26] E. Rabieinejad, A. Yazdinejad, A. Dehghantanha, and G. Srivastava, "Two-level privacy-preserving framework: Federated learning for attack detection in the consumer Internet of Things," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 4258–4265, Feb. 2024.
- [27] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artif. Intell. Statist.*, 2017, pp. 1273–1282.
- [28] N. Rieke et al., "The future of digital health with federated learning," *NPJ Digit. Med.*, vol. 3, no. 1, pp. 1–7, 2020.
- [29] A. Y. A. Amer, J.-M. Aerts, B. Vanrumste, and S. Luca, "A localized learning approach applied to human activity recognition," *IEEE Intell. Syst.*, vol. 36, no. 3, pp. 58–71, May/Jun. 2020.
- [30] R. Zuech, T. M. Khoshgoftaar, and R. Wald, "Intrusion detection and big heterogeneous data: A survey," *J. Big Data*, vol. 2, pp. 1–41, Feb. 2015.
- [31] T. R. Peltier, *Information Security Policies, Procedures, and Standards: Guidelines for Effective Information Security Management*. Boca Raton, FL, USA: CRC Press, 2016.
- [32] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How to backdoor federated learning," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2020, pp. 2938–2948.
- [33] V. Mothukuri, R. M. Parizi, S. Pouriyeh, Y. Huang, A. Dehghantanha, and G. Srivastava, "A survey on security and privacy of federated learning," *Future Gener. Comput. Syst.*, vol. 115, pp. 619–640, Feb. 2021.
- [34] S. E. A. Dadkhah, "CICIoMT2024: Attack vectors in healthcare devices—a multi-protocol dataset for assessing IoMT device security," *Internet Things*, vol. 28, Dec. 2024, Art. no. 101351.
- [35] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning," *IEEE Access*, vol. 10, pp. 40281–40306, 2022.
- [36] J. Zhang, J. Chen, D. Wu, B. Chen, and S. Yu, "Poisoning attack in federated learning using generative adversarial nets," in *Proc. 18th IEEE Int. Conf. Trust, Security Privacy Comput. Commun., 13th IEEE Int. Conf. Big Data Sci. Eng. (TrustCom/BigDataSE)*, 2019, pp. 374–380.